

PROJECT PRESENTATION

Fish Farm Management System

Developed by

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Introduction & Problems

Current System Issues

Currently, fish farm data (feed, expenses, sales) is maintained manually on paper registers.

- High risk of data loss.
- Calculation mistakes.
- Time-consuming manual updates.
- Lack of real-time data access.

Research Findings

Data collected through farmer interviews revealed critical operational bottlenecks:

- **Data Duplication:** Records are often repeated unnecessarily.
- **Sales Tracking:** Difficulty in monitoring sales history.
- **No Backup:** Critical business data has no secure online backup.

System Comparison

Feature	Dairy Farm	Poultry Farm	Crop Farm	Fish Farm (Proposed)
Main Focus	Health & Milk	Chickens & Eggs	Crop, Irrigation, Yield	Ponds, Feed, Sales, Expenses
Data Storage	Web + DB	Web + Local	IoT / Cloud	Mobile App (Firebase)
Users	Dairy Farmer	Poultry Owners	Crop Growers	Fish Farmer
Benefits	Automated Milk Logs	Health & Feed Control	Crop Monitoring	Easy Record Keeping
Output	Daily Milk Report	Egg Chart	Yield Graph	Profit/Loss Report

Project Proposal

Proposed Solution

A mobile application designed specifically for fish farmers to digitally record feed, sales, and expenses.

- **Automatic Calculations:** Instantly compute profit and loss.
- **Cloud Backup:** Ensures data security via the cloud.
- **User Friendly:** Simple interface for ease of use.

Waterfall Methodology

The project follows the standard Waterfall Model phases:

1. Requirement Gathering
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

System Analysis



Environment

Developed as an Android App using **Kotlin** or **Flutter**.



Database

Integrated with **Firebase** for real-time data sync and storage.



Connectivity

Works both **Online** and **Offline** with synchronization capability.

Scope: Efficient record maintenance for tanks, feed, and sales with automated reporting.

Functional Requirements

ID	Requirement	Description
FR1	Login/Register	User authentication through Firebase securely.
FR2	Add Ponds	Record pond number, fish quantity, and start date.
FR3	Add Feed	Store feed details: type, quantity, and cost.
FR4	Add Expense	Track expenses related to maintenance or operations.
FR5	Add Sales	Record fish sales including total amount and rate per kg.
FR6	View Report	Generate automated profit/loss summary reports.

Non-Functional Requirements



Usability

Interface must be simple for farmers.



Performance

Data loading within 2 seconds.



Security

Password protection & encryption.



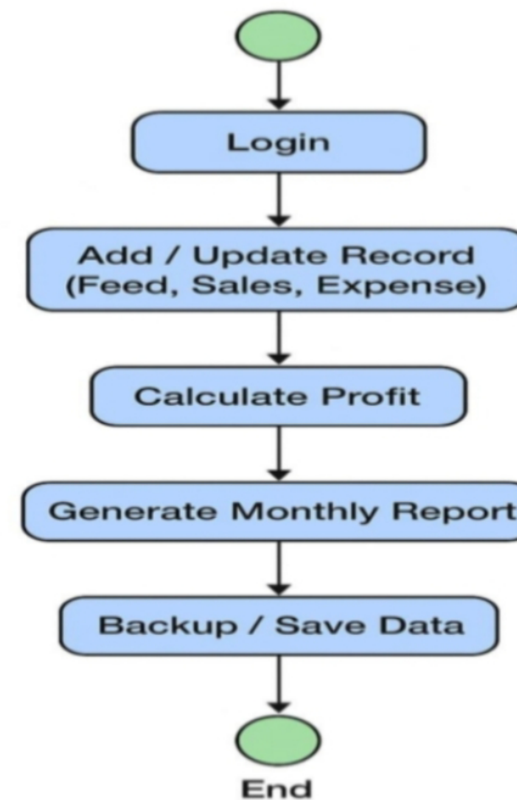
Reliability

Regular cloud backups.

Activity Diagram

This diagram illustrates the operational flow of the system:

- **Start:** User logs into the application.
- **Action:** User adds or updates records (Feed, Sales, Expenses).
- **Process:** System calculates profit based on inputs.
- **Output:** Monthly reports are generated and data is securely backed up.

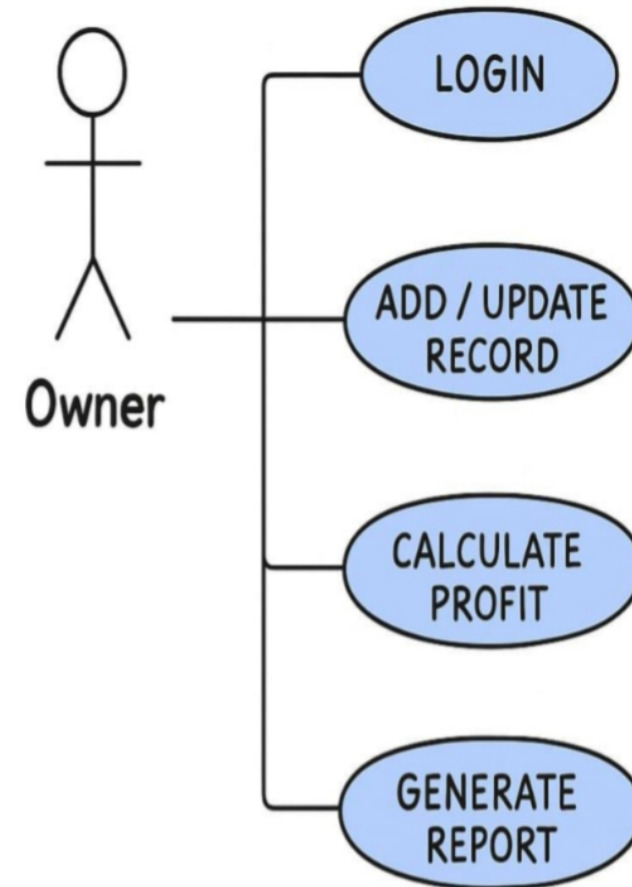


Use Case Diagram

Visualizing the interactions between the **Owner** and the system.

Key Use Cases:

- Login Authentication
- Add/Update Records
- Calculate Profit
- Generate Reports



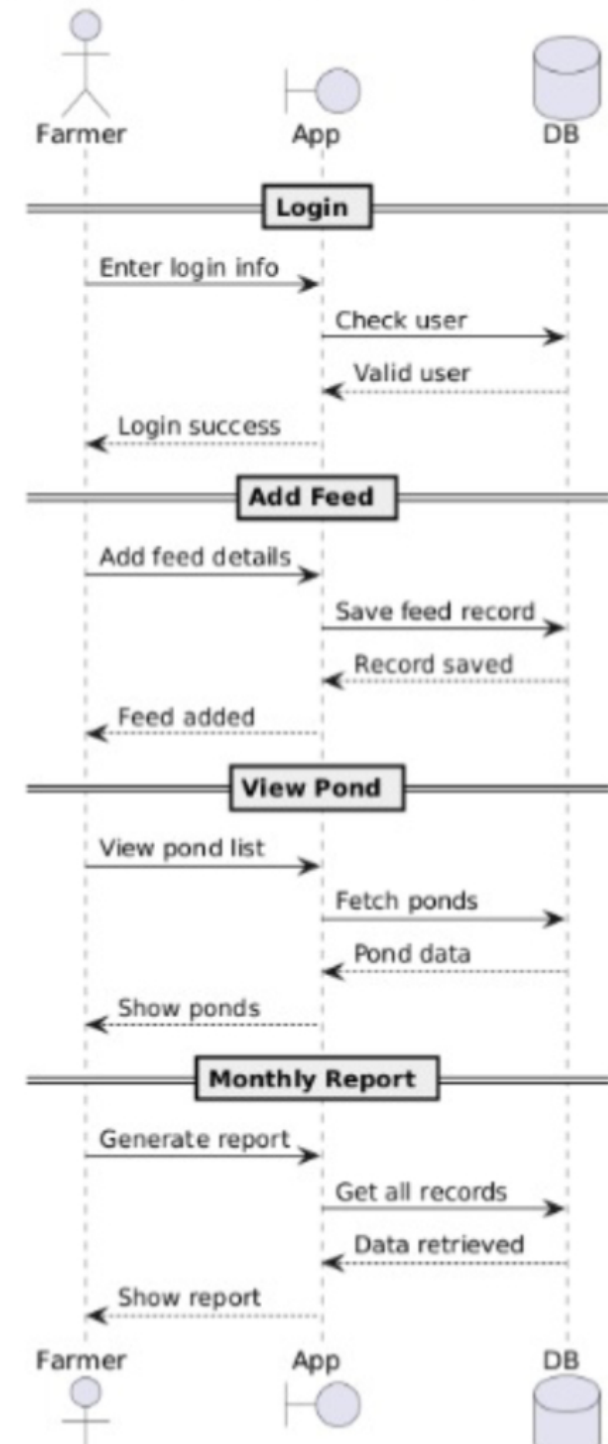
Sequence Diagram

Demonstrates the sequential logic flow for specific tasks, such as **Adding Feed** or **Viewing Reports**.

Flow: Farmer → Application → Database.

The system validates the user, processes the request, saves data to the database, and returns a confirmation to the interface.

Fish Farm Management System - Sequence Diagram

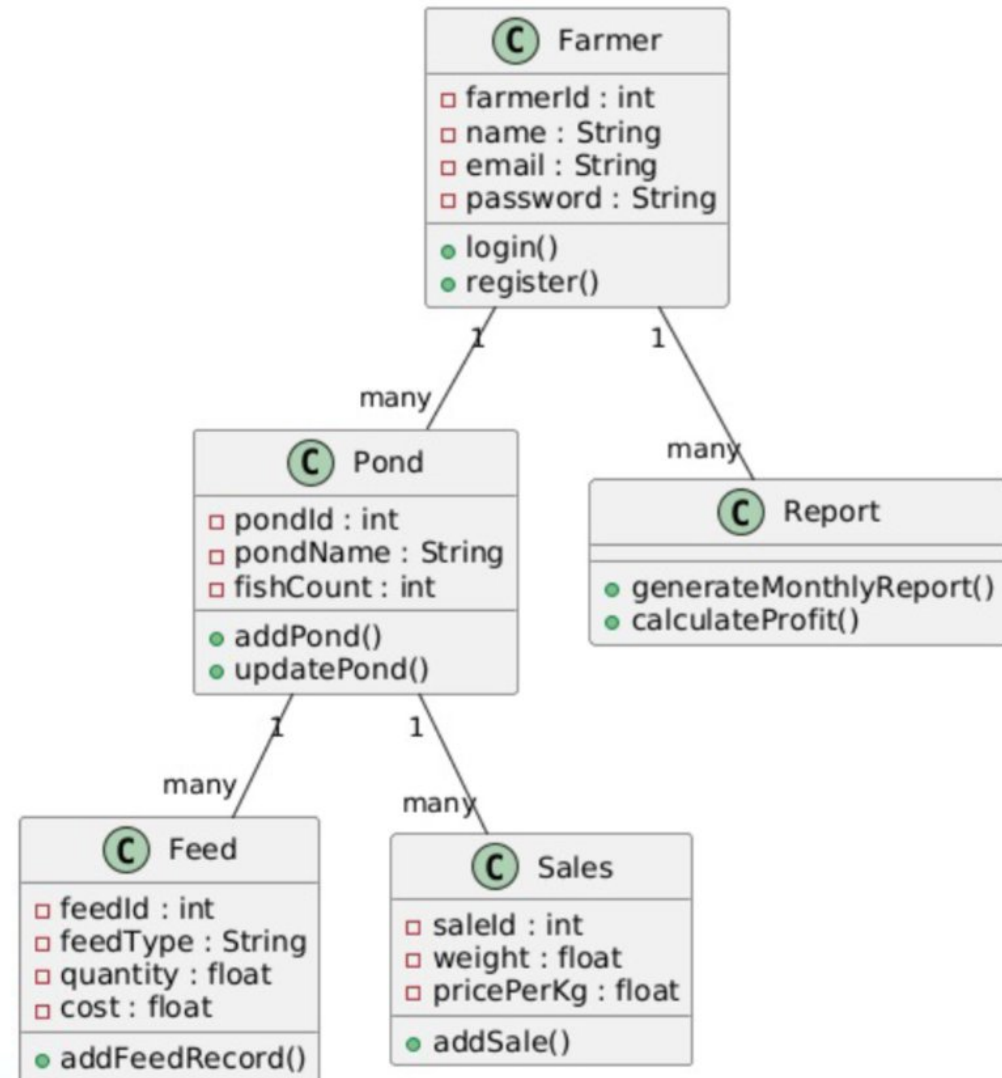


Class Diagram

Defines the structure of the system's data and classes.

- **Farmer:** Handles authentication (login, register).
- **Pond:** Manages pond details (ID, Name, Fish Count).
- **Feed & Sales:** Tracks inventory usage and revenue.
- **Report:** Methods for profit calculation and monthly summaries.

Fish Farm Management System - Class Diagram



Project Impact

The Fish Farm Management System successfully modernizes traditional farming methods.

- Simplifies record management.
- Prevents data loss.
- Saves valuable time.
- Provides a scalable platform for future IoT integration.

Hardware Requirements

Android Version	8.0 (Min) - 10.0+ (Rec)
RAM	2GB (Min) - 4GB+ (Rec)
Storage	200MB (Min) - 500MB+ (Rec)